

Vinh Ly

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Summary

Software Engineer with **18 years** designing and shipping large-scale distributed systems at Google, YouTube, and Microsoft. Built and led teams owning infrastructure serving billions of users, including video streaming pipelines at YouTube and marketplace systems at Whatnot. Expertise in **Kubernetes**, **GCP**, **Python**, **Go**, and **C++**. Drove platform reliability initiatives reducing incident response time by 40% at Google. Partners cross-functionally with product, data science, and security teams. Focus on AI-enabled cloud platforms and next-generation infrastructure. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Platforms: Kubernetes, GCP, AWS, Terraform, Docker

Languages: Python, Go, C++, Java, TypeScript, JavaScript, HTML, CSS, SQL

Infrastructure: CI/CD pipelines, service mesh, IaC

Systems: Microservices, distributed systems, real-time data pipelines, video streaming

Operations: On-call, incident management, capacity planning, SRE practices

Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Architect video streaming infrastructure handling 2B+ daily active users, designing fault-tolerant pipelines in **GCP** for content delivery optimization.
- Lead migration of legacy monolithic services to **Kubernetes**-based microservices, improving deployment frequency from weekly to multiple daily releases.
- Implement real-time analytics processing using **Python** and **Go** microservices, reducing latency for creator dashboard metrics by 60%.
- Collaborate with platform security teams to enforce zero-trust networking across video processing clusters.
- Mentor junior engineers on distributed systems design patterns and **GCP** best practices.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Designed marketplace platform infrastructure supporting live auctions with 500K+ concurrent users during peak events.
- Built **CI/CD pipelines** using **Terraform** and GitHub Actions, cutting infrastructure provisioning time from days to under 30 minutes.
- Established on-call rotation and incident response protocols, achieving 99.9% platform availability during first quarter post-launch.
- Led database query optimization initiative in **PostgreSQL** reducing p99 latency from 800ms to 150ms for product listings.
- Implemented event-driven architecture using Kafka for order processing, enabling real-time inventory updates across 50K+ daily listings.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Led team of 12 engineers building internal developer platform powering 200+ services across Search, Ads, and Cloud divisions.
- Reduced incident response time by 40% through implementation of automated runbook execution and **observability** dashboards in **Go**.
- Owned **Kubernetes** cluster fleet management for 50K+ nodes, designing multi-region failover strategies for critical infrastructure.

- Championed platform-wide adoption of **service mesh** (Istio) improving request routing reliability across **C++** and **Python** services.
- Defined capacity planning models using **Python** predictive analytics, optimizing cloud spend by \$2M annually.
- Partnered with security team to implement RBAC and secret management solutions across 500+ internal services.
- Conducted technical interviews and hiring for platform engineering org, scaling team from 5 to 15 engineers.

Software Engineer | Google | 02/2011 - 02/2015

- Developed backend systems for AdSense serving infrastructure handling 1M+ queries per second in **C++**.
- Built automated testing frameworks improving code coverage from 45% to 85% across core pipeline components.
- Designed **Python** scripts for log analysis and anomaly detection, identifying revenue-impacting bugs before customer escalation.
- Participated in on-call rotation for ad auction systems, resolving critical P1 incidents with 4-hour MTTR.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Designed developer tools and APIs for Visual Studio team, improving build times for enterprise customers by 25% using **C++** optimizations.
- Built automated testing infrastructure for Windows SDK components, reducing QA cycle time from 2 weeks to 3 days.
- Authored technical documentation and API guides adopted by 100K+ external developers.
- Collaborated with product management to define roadmap for developer productivity tools based on customer feedback.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Developed proof-of-concept features for Windows Mobile platform using **C++** and .NET framework.
- Presented internship project to 50+ engineers demonstrating memory optimization techniques for embedded systems.

Education

B.S. Honors in Computer Science

Virginia Tech | 2004 - 2007